

An Introduction to Universal Artificial Intelligence

Chapter 2.4: Bayesian Probability Theory

Based on Hutter, Quarel & Catt (2024)

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From frequentist statistics to Bayesian statistics

The previous section (2.3) covered *frequentist* statistics: powerful and elegant estimators such as the maximum likelihood estimator (MLE), but with no single unifying principle, and with the uncertainty about the true probability measure handled only indirectly (e.g. via confidence intervals, which are routinely misinterpreted).

The Bayesian idea

In Bayesian statistics, uncertainty about the underlying probability measure is itself modelled *by probabilities*. Starting from a *prior* w_θ (belief = epistemic probability) over the unknown parameter $\theta \in \Theta$, Bayes' rule is the general and optimal learning rule that uniquely determines how to update the prior to a *posterior* belief probability once new data (evidence) arrives.

We first recall Bayes' rule for events, then specialise it to a simple binary i.i.d. setting, which leads to the famous *Laplace rule*. Later chapters extend this machinery to much larger, in fact *universal*, model classes and priors (Solomonoff induction, AIXI).

Example 2.4.1: Estimating the lottery-winning probability

Setup

Consider buying your very first lottery ticket, and you get lucky and win on the first try. Let θ be the true (unknown) probability of winning with a single ticket.

Lacking any prior knowledge about lotteries, the maximum likelihood estimator (MLE) would conclude from this single data point “win” that $\hat{\theta}_{\text{MLE}} = 1$: i.e. that you are *guaranteed* to win the lottery every single time you play.

Why this is implausible. We already possess strong prior knowledge about lotteries: out of the many tickets sold, only a few are winners, so winning should be a very rare event. A frequentist estimator that ignores this prior knowledge and treats every parameter value as equally consistent with the data can produce absurdly overconfident conclusions when the sample size is tiny.

Coin-flip analogy. If we are estimating the bias θ (theta) of a coin, we expect an ordinary coin to have $\theta \approx \frac{1}{2}$. After observing the coin land at least once on heads and at least once on tails, we can already rule out the extreme options $\theta = 0$ or $\theta = 1$ – something the naive MLE cannot do robustly from very little data.

Fixing the problem: priors and posteriors

- A frequentist approach *can* converge to the truth in the limit of a lot of data, but for *small* amounts of data we should also incorporate current prior domain knowledge about the parameter of interest into the estimation procedure.
- The distribution over the set of parameters given the domain knowledge, *before* seeing any data, is called the *prior*.
- **Bayes' Rule** gives a systematic method for turning the prior into a new distribution after having received the data, called the *posterior*.

Roadmap for this section

- 1 **2.4.1** Bayes' Theorem for events, and the different interpretations of "probability."
- 2 **2.4.2** Bayes estimation and prediction for a parameter θ : joint distribution, mixture/evidence, posterior, predictive distribution.
- 3 **2.4.3** The Laplace rule: a fully worked binary i.i.d. example, the Beta distribution, the Jeffreys/KT prior, and the mean/variance of the posterior.

Bayes' celebrated theorem

Bayes' theorem allows the conditional probability $P(A | B)$ to be computed in terms of $P(A)$, $P(B)$, and $P(B | A)$.

Theorem 2.4.2 (Bayes' theorem)

Let $D \subseteq \Omega$ be an event with $P(D) > 0$ and $(H_i)_{i \in I}$ be a countable partition of Ω (i.e. $\forall i \neq j : H_i \cap H_j = \emptyset$ and $\bigcup_i H_i = \Omega$). Then

$$P(H_i | D) = \frac{P(D | H_i) P(H_i)}{P(D)} = \frac{P(D | H_i) P(H_i)}{\sum_{j \in I} P(D | H_j) P(H_j)}.$$

Plain-English reading of every symbol. Ω is the whole sample space (“everything that could happen”); the sets H_i are mutually exclusive *hypotheses* that together cover every possibility (a *partition*); D is the observed *data* or *evidence*; $P(H_i)$ is the *prior* belief in hypothesis H_i (before seeing D); $P(D | H_i)$ is the *likelihood* of the data under hypothesis H_i ; and $P(H_i | D)$ is the *posterior* belief in H_i after observing D . The key quantity $P(D) = \sum_j P(D | H_j) P(H_j)$ is called the *Bayesian evidence*, the *marginal distribution*, or the *mixture distribution*.

Interpretation: what does Bayes' theorem give us? I

The intuition behind Bayes' theorem is that it provides a method of *updating* (belief) probabilities based on evidence. Say H_1 is the hypothesis that a person has a particular disease, and H_0 that they do not. Before any clinical test, the doctor holds a subjective belief $P(H_i)$, e.g. based on symptoms and demographics. After the test result D arrives, this belief updates to $P(H_i | D)$, which may increase or decrease the confidence in H_i depending on whether D is evidence for or against it.

Mathematically, the Bayesian “sample” space Ω is a product of a data (frequentist) sample space Ω_D and a hypothesis space Ω_H . A hypothesis H_i is an event of the form $\Omega_D \times \{\tilde{H}_i\} \subset \Omega$; an observation D is an event of the form $\tilde{D} \times \Omega_H \subset \Omega$. Usually $\tilde{H}_i \in \Omega_H$ and $\tilde{D} \subset \Omega_D$ are what get specified, with the tilde dropped and the lifting to the product space implicit.

Different people can hold different confidences in the same events: the patient and the doctor generally have different priors $P(H_i)$ (the doctor has more experience judging base rates, the patient's belief may be driven more by fear or wishful thinking), and hence, even after both learn about D , they will typically hold different posteriors too.

Interpretation: what does Bayes' theorem give us? II

Four interpretations of “probability.”

- **Frequentist:** Probabilities are the relative frequency of an event occurring. In a sequence of n i.i.d. trials, if an event occurs $k(n)$ times, its relative frequency is $k(n)/n$, and the probability of the event is defined as the limit $\lim_{n \rightarrow \infty} k(n)/n$. Frequentists model $P(D | H_i)$ but assign *neither* a prior nor a posterior probability to H_i .
- **Objectivist:** Probabilities are real aspects of the world (“true” probabilities). Outcomes are not deterministic but involve chaotic or random physical processes. They should be consistent with limiting frequencies for i.i.d. experiments, but are not limited to that case – e.g. the probability of a volcanic eruption in a specific century is a unique, non-i.i.d., stochastic-process probability.
- **Subjectivist:** Probabilities are merely subjective, and represent a *particular individual's* confidence or degree of belief, called *epistemic* probability. Subjective probabilities depend on who holds them (doctor vs. patient) and on what extra information is available.
- **Bayesian:** A Bayesian assigns a prior probability over H_i and uses a (frequentist) likelihood model $P(D | H_i)$ to update the prior $P(H_i)$ to a posterior $P(H_i | D)$ after observing evidence D . The prior may be *subjective* (e.g. from expert elicitation) or *objective* (like the Jeffreys prior below), or *universal* (Section 3.7) – though this trichotomy is a gross over-simplification.

Interpretation: what does Bayes' theorem give us? III

Remarkably, *all* of these interpretations satisfy the very same Kolmogorov axioms of probability – they differ only in what probability is taken to *mean*, not in the mathematics governing it.

Example 2.4.3: Blood type testing

Setup

Blood type O-negative (O^-) can be donated to anyone, and 7% of the population has it: $P(H^-) = 0.07$ where H^- is “Person A has blood type O^- ” and $H^+ = \overline{H^-}$. A test is 95% accurate: if you truly have (do not have) O^- blood, it returns a false negative (positive) with probability 5%. The test on a random person comes back *positive* for O^- . How confident should we be that this person *actually* has O^- blood?

Let D^- be the event “the test says O^- .” The likelihoods are $P(D^- | H^-) = P(D^+ | H^+) = 0.95$ (correct results) and $P(D^- | H^+) = P(D^+ | H^-) = 0.05$ (errors). Applying Bayes’ Theorem 2.4.2 with hypothesis space $\Omega_H = \{H^-, H^+\}$:

$$P(H^- | D^-) = \frac{P(D^- | H^-) P(H^-)}{P(D^- | H^-) P(H^-) + P(D^- | H^+) P(H^+)} = \frac{0.95 \times 0.07}{0.95 \times 0.07 + 0.05 \times 0.93} \approx 59\%.$$

Every symbol: the numerator is (probability test is correct) $\times$ (prior probability of actually having O^-); the denominator additionally sums in the probability that the test falsely reports O^- for an H^+ person.

Blood type testing: why the answer feels counter-intuitive

A symmetric calculation for a *negative* test result gives

$$P(H^- | D^+) = \frac{P(D^+ | H^-) P(H^-)}{P(D^+ | H^-) P(H^-) + P(D^+ | H^+) P(H^+)} = \frac{0.05 \times 0.07}{0.05 \times 0.07 + 0.95 \times 0.93} \approx 0.4\%.$$

Interpretation. Despite the test having a relatively high success probability (95%), the probability of *actually* having O^- blood given a positive test is only $\approx 59\%$. This is due to the small *base rate* (prior probability) of O^- blood: since the probability of a false positive (5%) and the probability of truly having O^- blood (7%) are of similar magnitude, roughly half the positive test results come from the test being wrong and half from genuinely O^- patients. Conversely, since non- O^- blood is so common, a *negative* result is very strong evidence of not having O^- blood, since that would require *both* an unlikely event (having O^-) *and* the reliable test failing.

Take-away

Always weigh the likelihood ratio against the *prior odds* $P(H^-)/P(H^+)$ – an accurate test can still leave you quite uncertain when the condition being tested for is rare.

Python: verifying the blood-type posterior

```
# Exact re-derivation of Example 2.4.3 by direct computation.
P_Hminus = 0.07          # prior  $P(H^-)$ : 7% of population is O-negative
P_Hplus  = 1 - P_Hminus  # prior  $P(H^+)$ 

# Likelihoods: test is 95% accurate either way
P_Dminus_given_Hminus = 0.95  #  $P(D^-|H^-)$ : true positive rate
P_Dminus_given_Hplus  = 0.05  #  $P(D^-|H^+)$ : false positive rate
P_Dplus_given_Hminus  = 0.05  #  $P(D^+|H^-)$ : false negative rate
P_Dplus_given_Hplus   = 0.95  #  $P(D^+|H^+)$ : true negative rate

# Bayes' theorem: posterior = likelihood * prior / evidence
num_pos = P_Dminus_given_Hminus * P_Hminus
den_pos = num_pos + P_Dminus_given_Hplus * P_Hplus
post_Hminus_given_Dminus = num_pos / den_pos

num_neg = P_Dplus_given_Hminus * P_Hminus
den_neg = num_neg + P_Dplus_given_Hplus * P_Hplus
post_Hminus_given_Dplus = num_neg / den_neg

print(f"P(H- | test positive) = {post_Hminus_given_Dminus:.4f}") # ~0.5891
print(f"P(H- | test negative) = {post_Hminus_given_Dplus:.4f}")  # ~0.0038
```

Running this script reproduces exactly $\approx 59\%$ and $\approx 0.4\%$, confirming the hand calculation above and showing how directly Bayes' theorem translates into three lines of arithmetic.

From events to parameters: modelling uncertainty about θ

We now express our belief over plausible and implausible values of $\theta \in \Theta$ as a *density function* $w(\theta)$, called the *prior*.

- In *frequentist* statistics, θ is **not** a random variable: the true value of the parameter (e.g. the bias of a coin) is a fixed, though unknown, constant. Frequentists *can* express uncertainty in θ (e.g. via confidence intervals), but the interpretation is convoluted and often confused even by statisticians.
- *Bayesians* instead model uncertainty about θ in exactly the same way as for observations X : they *treat θ as a random variable* and assign a (subjective) prior distribution $w(\theta)$ over it, encoding our prior beliefs about the relative plausibility of values of $\theta \in \Theta$.

This requires enlarging the sample space to $\Omega'' = \Omega \times \Omega'$ to accommodate θ , where $X(\omega)$ becomes $X(\omega'')$. Typically $\Omega' = \Theta$, and with some abuse of notation $\theta(\omega') = \omega'$. Mathematically θ is now a random variable and Θ is part of the sample space – even though most Bayesians would *not* literally interpret θ as “random” or “sampled”; the clash in nomenclature is more than compensated by the mathematical convenience this unified treatment offers.

Joint, marginal, and posterior distributions

The probability distribution $p_X(x; \theta)$ for observed random variables X can be viewed as a conditional distribution on X given θ , formalised by defining the *joint* distribution

$$p(x, \theta) := p_X(x; \theta) w(\theta).$$

Every symbol: x is a possible value of the data, θ (theta) is the unknown parameter, $p_X(x; \theta)$ is the model's likelihood of x given θ , and $w(\theta)$ is the prior density over θ . Then $p(x | \theta) \equiv p(x, \theta)/w(\theta) = p_X(x; \theta)$ whenever $w(\theta) > 0$.

The mixture (marginal / Bayesian-evidence) distribution

$$p(x_1, \dots, x_n) = \int_{\Theta} p(x_1, \dots, x_n | \vartheta) w(\vartheta) d\vartheta.$$

This is the probability of the data *averaged over all values of θ* , weighted by the prior; ϑ (a variant form of theta) is just the integration (dummy) variable ranging over Θ . This quantity is central to Bayesian statistics: **as a rule of thumb, if the mixture can be efficiently computed, then everything else of interest can be efficiently computed too.**

Bayes' rule for parameters: prior $\rightarrow$ posterior

A continuous version of Bayes' Theorem 2.4.2 updates the prior belief about θ with observations $x_1, \dots, x_n$ to arrive at the *posterior* belief distribution

Posterior distribution

$$w(\theta \mid x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n \mid \theta) w(\theta)}{p(x_1, \dots, x_n)}.$$

Every symbol: the numerator multiplies the *likelihood* of the observed data given θ by the *prior* density at θ ; the denominator is the mixture distribution (evidence) from the previous slide, which normalises the result back into a valid density over θ . This posterior combines two sources of information: our prior beliefs about what θ should be, and any evidence provided by the observed data. Bayes' rule *uniquely* determines the posterior given a prior – i.e. it is the one and only consistent way to update beliefs based on evidence.

Predicting new data from the posterior

Often estimating θ is merely a means to an end: what we ultimately want is to make *predictions*.

Predictive distribution (2.4.4)

$$p(x_{n+1} \mid x_1, \dots, x_n) = \int_{\Theta} p(x_{n+1} \mid \vartheta) w(\vartheta \mid x_1, \dots, x_n) d\vartheta = \frac{p(x_1, \dots, x_{n+1})}{p(x_1, \dots, x_n)}.$$

Every symbol: x_{n+1} is the next, not-yet-observed data point; $w(\vartheta \mid x_1, \dots, x_n)$ is the posterior density over θ from the previous slide; and the integral averages the model's prediction $p(x_{n+1} \mid \vartheta)$ over every possible value of ϑ , weighted by how plausible the posterior currently considers that value. The predictive distribution therefore *averages the model's prediction over the posterior uncertainty in θ* , rather than committing to a single point estimate of θ and predicting from that alone – this is what makes Bayesian prediction naturally “humble” about parameter uncertainty.

Setting up the running example: a biased coin

Section 2.4.2 explains *how* to incorporate a prior, but is agnostic about *where the prior comes from*. In practice, the choice of prior can greatly affect the estimate when little data is available, though with enough data the posterior concentrates on the true parameter for “sensible” prior choices. Choosing a canonical prior over a *function space* that respects Occam’s razor is a major focus of later chapters of the book (universal priors); here we work out the simplest possible case fully.

Derivation setup

Suppose we have a biased coin governed by a Bernoulli process with unknown parameter θ : we observe a sequence of coin flips $x_1, x_2, \dots \sim \text{Bern}(\theta)$, and would like to estimate θ from the sequence. Applying Bayes’ Theorem 2.4.2 gives the posterior

$$w(\theta \mid x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n \mid \theta) w(\theta)}{\int_{\Theta} p(x_1, \dots, x_n \mid \vartheta) w(\vartheta) d\vartheta} = \frac{w(\theta) \prod_{i=1}^n \text{Bern}(x_i; \theta)}{\int_{\Theta} w(\vartheta) \prod_{i=1}^n \text{Bern}(x_i; \vartheta) d\vartheta}. \quad (2.4.5)$$

Since $\text{Bern}(x; \theta) = \theta^x (1 - \theta)^{1-x}$, the product $\prod_{i=1}^n \text{Bern}(x_i; \theta) = \theta^b (1 - \theta)^a$, where a is the number of zeros and b the number of ones in $x_1, \dots, x_n$.

The Beta distribution

Definition (2.4.6): the Beta distribution

$$\text{Beta}(\theta; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}, \quad \text{where } B(\alpha, \beta) = \int_0^1 \theta^{\alpha-1} (1 - \theta)^{\beta-1} d\theta$$

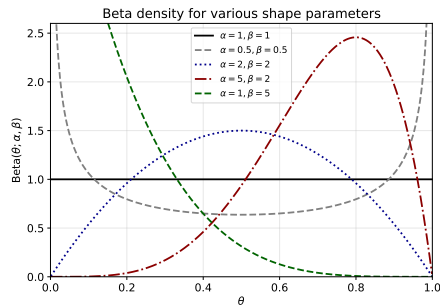
is the *beta function*, and α (alpha), β (beta) > 0 are the two shape parameters of the distribution.

If we have no prior information about the coin, choosing a **uniform** distribution over $[0, 1]$ for our prior belief in θ is reasonable. This should **not** be interpreted as a “lack of a prior belief” – it in fact expresses a prior belief of *indifference*: every value in $[0, 1]$ is deemed *a priori* as likely as any other. Using $w(\theta) = 1$ for $0 \leq \theta \leq 1$ turns (2.4.5) into

$$w(\theta \mid x_1, \dots, x_n) = \frac{\theta^b (1 - \theta)^a}{\int_0^1 \vartheta^b (1 - \vartheta)^a d\vartheta} = \frac{\theta^b (1 - \theta)^a}{B(b + 1, a + 1)} = \text{Beta}(\theta; b + 1, a + 1).$$

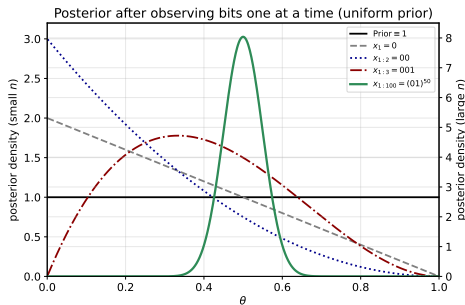
So starting from a uniform prior, observing b ones and a zeros turns the prior into a $\text{Beta}(b + 1, a + 1)$ posterior – the counts a, b simply shift the Beta’s shape parameters.

Figure: the Beta density for various shape parameters



As α, β (alpha, beta) increase together and stay equal, the density concentrates more sharply around $\theta = \frac{1}{2}$ (e.g. the dotted curve $\alpha = \beta = 2$); when $\alpha \neq \beta$, the mode shifts towards 1 or 0 (e.g. the dash-dot curve $\alpha = 5, \beta = 2$ favours large θ); and when $0 < \alpha = \beta < 1$ (dashed curve $\alpha = \beta = 0.5$), probability mass piles up near the two extremes 0 and 1 instead of the middle.

Figure: watching the posterior update, one bit at a time



Starting from a uniform prior ($n = 0$, solid black line), the sequence $x_{1:3} = 001$ is observed one bit at a time: the posterior shifts more probability mass towards *smaller* θ after each 0, then shifts back slightly once the 1 is observed. The green curve (right-hand axis) shows the far sharper posterior after 100 flips of alternating 0s and 1s, $x_{1:100} = (01)^{50}$: with far more data the posterior concentrates tightly around $\theta = \frac{1}{2}$, exactly as expected for a fair coin.

Predicting the next flip: deriving the Laplace rule

Given the posterior over θ , we use the predictive distribution (2.4.4) to predict x_{n+1} . Since $X_{n+1} \sim \text{Bern}(\theta)$ means $P(X_{n+1} = 1 \mid \theta) = \theta$,

$$P(X_{n+1} = 1 \mid x_1, \dots, x_n) = \int_{\Theta} \theta \cdot \frac{\theta^b (1 - \theta)^a}{B(b + 1, a + 1)} d\theta = \frac{\int_0^1 \theta^{b+1} (1 - \theta)^a d\theta}{B(b + 1, a + 1)} = \frac{B(b + 2, a + 1)}{B(b + 1, a + 1)}.$$

The Gamma function. We simplify this using the *Gamma function*

$$\Gamma(k) := \int_0^\infty x^{k-1} e^{-x} dx \quad (2.4.7),$$

a continuous generalisation of the factorial, satisfying $\Gamma(k) = (k - 1)!$ for integer $k \geq 1$ and $\Gamma(z + 1) = z\Gamma(z)$ for all $z \in \mathbb{R}$ where $\Gamma(z)$ is defined. The Beta function satisfies, for any $a, b > 0$,

$$B(a, b) = B(b, a) = \int_0^1 \theta^{b-1} (1 - \theta)^{a-1} d\theta = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a + b)}.$$

Substituting this identity into $B(b + 2, a + 1)/B(b + 1, a + 1)$ and cancelling common Gamma-function factors (using $\Gamma(z + 1) = z\Gamma(z)$ repeatedly) yields, after simplification,

$$P(X_{n+1} = 1 \mid x_1, \dots, x_n) = \frac{b + 1}{a + b + 2} = \frac{1}{n + 2} \left(1 + \sum_{i=1}^n x_i \right).$$

Theorem 2.4.8: the Laplace rule

Theorem 2.4.8 (Laplace rule)

For i.i.d. $X_i \sim \text{Bern}(\theta)$ and a uniform prior $\theta \in [0, 1]$,

$$P(X_{n+1} = 1 \mid X_1 = x_1, \dots, X_n = x_n) = \frac{k+1}{n+2}, \quad \text{where } k = |\{i \leq n : x_i = 1\}| = \sum_{i=1}^n x_i.$$

Every symbol: n is the number of coin flips observed so far, k is how many of them came up “1,” and the formula $(k+1)/(n+2)$ is the probability the estimator assigns to the *next* flip being a 1. This particular choice of (uniform) prior is called the *indifference rule* or *Laplace rule*, and the resulting estimator is the *Laplace estimator*.

The sunrise problem. If the sun has risen every day for all $1 \leq i \leq n$ (so $x_i = 1$ always), the Laplace rule gives the estimate $\frac{n+1}{n+2}$ that the sun will rise tomorrow – a much less overconfident answer than the naive MLE’s certainty of 1, yet it still approaches 1 as more days of evidence accumulate.

Python: Laplace rule in action

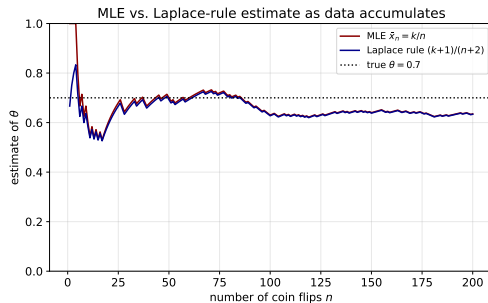
```
def laplace_rule(sequence):
    """P(next flip = 1 | sequence), Theorem 2.4.8."""
    n = len(sequence)
    k = sum(sequence)          # number of ones observed
    return (k + 1) / (n + 2)

def mle(sequence):
    """Naive frequentist estimate: relative frequency of ones."""
    n = len(sequence)
    return None if n == 0 else sum(sequence) / n # undefined @ n=0

# Example 2.4.1: a single winning lottery ticket
print("MLE after one win:      ", mle([1]))          # 1.0 (overconfident!)
print("Laplace after one win:", laplace_rule([1]))   # 0.667
# Sunrise problem: n=10000 days, sun rose every day
seq = [1] * 10000
print("MLE (sunrise, n=10000):  ", mle(seq))          # 1.0
print("Laplace (sunrise, n=10000):", laplace_rule(seq)) # 0.99980002
# Coin-flip sequence x_{1:3} = 0,0,1 from Figure 2.11
print("Laplace after 001:", laplace_rule([0, 0, 1])) # 0.4
```

The Laplace rule never outputs exactly 0 or 1 for finite n , unlike the MLE, reflecting the Bayesian's built-in "humility" from the uniform prior.

Figure: MLE versus the Laplace-rule estimate



A simulated sequence of 200 coin flips with true bias $\theta = 0.7$ (dotted black line). The MLE (red) is noisier for small n and can briefly touch 0 or 1; the Laplace-rule estimate (blue) is systematically pulled towards $\frac{1}{2}$ for small n (reflecting the uniform prior's “pull”) but the two curves converge as n grows, since the influence of any fixed prior vanishes as evidence accumulates.

The general Beta prior

More generally, suppose we choose an arbitrary Beta distribution as our prior, $w(\theta) = \text{Beta}(\theta; \alpha, \beta)$ for some $\alpha, \beta > 0$ (the uniform prior is the special case $\alpha = \beta = 1$, since $\text{Beta}(\theta; 1, 1) = 1$). Repeating the same calculation as before yields

Posterior and predictive distribution, general Beta prior

$$w(\theta \mid x_1, \dots, x_n; \alpha, \beta) = \text{Beta}(\theta; b + \alpha, a + \beta) \quad (2.4.9)$$

$$P(X_{n+1} = 1 \mid x_1, \dots, x_n; \alpha, \beta) = \frac{b + \beta}{a + b + \alpha + \beta} = \frac{\beta + \sum_{i=1}^n x_i}{n + \alpha + \beta}. \quad (2.4.10)$$

Pseudocounts. The parameters α and β can be thought of as *hypothetical prior counts* of zeros and ones observed before the experiment even began (as if we had been told the outcome of some earlier, hypothetical experiment with the coin). Because they act like extra, imaginary observations that smooth out the rapid changes seen when very little real data is available, α and β are called *pseudocounts*, and they act as a *regularizer*.

Choosing α and β

If we have no information about whether a 0 or a 1 is more likely, *a priori* it seems sensible to choose $\alpha = \beta$, eliminating one free parameter – but we still need to pick the common value.

- $\alpha = \beta = 1$: recovers the uniform (indifference) prior.
- $\alpha = \beta > 1$: weights the prior more heavily towards $\theta \approx \frac{1}{2}$, and is *unlikely* to move towards the extremes 0 or 1 without a lot of contrary evidence. Appropriate when we have a strong prior belief that a coin is fair: with large $\alpha = \beta$, only a sufficiently large number of observations would persuade us the coin is unfair.
- $0 < \alpha = \beta < 1$: the opposite case – a prior that is *more* biased towards the extremes, expressing a prior assumption that the coin is likely to be unfair. In the extreme limit $\alpha = \beta \rightarrow 0$, the distribution degenerates into two point masses at 0 and 1, i.e. a prior assumption that the coin's outcome is fully *deterministic*.

The Jeffreys prior

One principled approach for choosing $\alpha = \beta$ is to require the prior to be *invariant under a change of coordinates* for the parameter θ .

Definition 2.4.11 (Jeffreys prior)

The Jeffreys prior is a prior with density function proportional to the square root of the Fisher information:

$$w(\theta) \propto \sqrt{\mathcal{I}(\theta)}.$$

Here $\mathcal{I}(\theta)$ (the calligraphic letter I) is the *Fisher information* introduced in Section 2.3: it measures how much information each sample carries, on average, about θ .

We compute $\mathcal{I}(\theta)$ for $\text{Bern}(x; \theta)$:

$$\mathcal{I}(\theta) = -\mathbf{E}_{\theta} \left[\frac{\partial^2}{\partial \theta^2} \ln \text{Bern}(X; \theta) \right] = \mathbf{E}_{\theta} \left[\frac{X}{\theta^2} + \frac{1-X}{(1-\theta)^2} \right] = \frac{\theta}{\theta^2} + \frac{1-\theta}{(1-\theta)^2} = \frac{1}{\theta(1-\theta)},$$

using linearity of expectation and $\mathbf{E}_{\theta}[X] = \theta$. Hence $w(\theta) \propto 1/\sqrt{\theta(1-\theta)}$, which is exactly a Beta distribution with $\alpha = \beta = \frac{1}{2}$.

The KT estimator, and Jeffreys vs. uniform priors

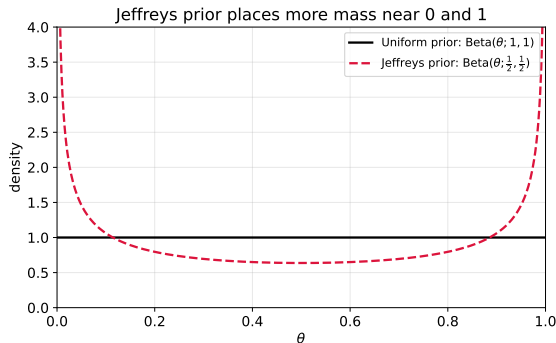
The KT estimator

The estimator resulting from the Jeffreys prior $\text{Beta}(\theta; \frac{1}{2}, \frac{1}{2})$ is called the *KT estimator* (Krichevsky–Trofimov), explored further in Section 4.1. While this choice is invariant under reparametrisation and minimax optimal for a binary alphabet, other choices of α, β can perform better for large, non-binary alphabets.

From (2.4.10) with $\alpha = \beta = \frac{1}{2}$, the KT predictive probability is

$$P(X_{n+1} = 1 \mid x_1, \dots, x_n) = \frac{b + \frac{1}{2}}{n + 1}.$$

As the figure shows, the Jeffreys/KT prior (dashed red) places *more* density near the extremes $\theta = 0, 1$ than the flat uniform prior (solid black), reflecting the fact that a $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ prior is slightly more open to a strongly biased coin than the indifference prior is.



Python: KT estimator vs. Laplace estimator

```
def beta_predictive(sequence, alpha, beta):  
    """P(next flip = 1 | sequence), general Beta(alpha,beta) prior, (2.4.10)."""  
    n = len(sequence)  
    b = sum(sequence)           # number of ones  
    return (b + beta) / (n + alpha + beta)  
  
seq = [0, 0, 1]  # x_{1:3} = 001, as in Figure 2.11 of the book  
  
laplace_pred = beta_predictive(seq, alpha=1, beta=1)      # uniform prior  
kt_pred      = beta_predictive(seq, alpha=0.5, beta=0.5) # Jeffreys/KT prior  
  
print(f"Laplace rule P(x4=1|001) = {laplace_pred:.4f}") # 0.4000  
print(f"KT estimator P(x4=1|001) = {kt_pred:.4f}")      # 0.3750  
  
# Strong-fair-coin prior: alpha=beta=20 (pseudocounts of 20 heads, 20 tails)  
fair_pred = beta_predictive(seq, alpha=20, beta=20)  
print(f"Strong-fair-coin prior P(x4=1|001) = {fair_pred:.4f}") # 0.4767
```

With only three observations, the strong “fair-coin” prior ($\alpha = \beta = 20$) barely moves away from $\frac{1}{2}$, while the KT and Laplace estimators react much more quickly to the data – exactly the behaviour described by the concentration-parameter discussion below.

Decomposition: Bayes estimator as a convex combination

Equation (2.4.10) admits a clean decomposition:

Decomposition and concentration (2.4.12)

$$P(X_{n+1} = 1 \mid x_1, \dots, x_n; \alpha, \beta) = \lambda \bar{x} + (1 - \lambda) \left(\frac{\beta}{\alpha + \beta} \right), \quad \lambda = \frac{n}{n + \alpha + \beta}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Every symbol: $\bar{x}$ (x-bar) is the plain relative frequency of ones – i.e. the MLE, based solely on the data with no prior knowledge; the term $\beta/(\alpha + \beta)$ is the *mean of the prior* $\text{Beta}(\theta; \alpha, \beta)$, i.e. the “pure prior” estimate returned when no data at all is available; and $\lambda \in [0, 1]$ (lambda) is a mixing weight. So the Bayes predictive estimator is exactly a **convex combination of the MLE and the pure-prior estimate**. Since $\lambda \rightarrow 1$ as $n \rightarrow \infty$ (for fixed α, β), the influence of the prior fades away as more data arrives – and because $\alpha + \beta$ controls how much weight the prior estimate retains for any fixed n , the quantity $\alpha + \beta$ is called the *concentration parameter*.

Posterior mean and variance: confidence in θ

As for the posterior $\text{Beta}(\theta; b + \alpha, a + \beta)$ itself, its mean and variance are

Posterior mean and variance (2.4.13)–(2.4.14)

$$\mathbf{E}[\theta \mid x_1, \dots, x_n] = \frac{b + \beta}{n + \alpha + \beta}, \quad \mathbf{Var}[\theta \mid x_1, \dots, x_n] = \frac{(a + \alpha)(b + \beta)}{(a + \alpha + b + \beta)^2(a + \alpha + b + \beta + 1)}.$$

Every symbol: a, b are again the number of zeros and ones observed; the mean is exactly the predictive probability derived above; and the variance shrinks as either the amount of data $n = a + b$ grows or the concentration $\alpha + \beta$ (i.e. prior strength) grows.

Take-away

The posterior has *low variance* (hence *high confidence* in θ) whenever **at least one of n or $\alpha + \beta$ is large**. For $\alpha + \beta \gg n$, the posterior concentrates around the prior belief $\beta/(\alpha + \beta)$; for $n \gg \alpha + \beta$, the posterior concentrates around the relative frequency b/n of ones – matching the intuition that with enough real data, the data “overrules” whatever prior we started with.

Chapter 2.4 summary

- **Bayes' theorem** (Theorem 2.4.2) turns a prior $P(H_i)$ and a likelihood $P(D | H_i)$ into a posterior $P(H_i | D)$, normalised by the evidence $P(D)$.
- Probability has (at least) four common interpretations – frequentist, objectivist, subjectivist, Bayesian – which differ in *meaning* but all obey the same Kolmogorov axioms.
- Treating an unknown parameter θ as a random variable with prior $w(\theta)$ lets us apply Bayes' rule to *parameter estimation*: joint $p(x, \theta)$, mixture/evidence $p(x_1, \dots, x_n)$, posterior $w(\theta | x_1, \dots, x_n)$, and predictive distribution $p(x_{n+1} | x_1, \dots, x_n)$.
- For i.i.d. Bernoulli data and a $\text{Beta}(\alpha, \beta)$ prior, the posterior stays Beta (a *conjugate* family), and the predictive probability of the next 1 is $(b + \beta)/(n + \alpha + \beta)$.
- $\alpha = \beta = 1$ gives the classical **Laplace rule** $(k + 1)/(n + 2)$; $\alpha = \beta = \frac{1}{2}$ (the **Jeffreys prior**) gives the **KT estimator**, invariant under reparametrisation.
- The Bayes predictive estimator is always a *convex combination* of the MLE and the pure-prior mean, with the *concentration parameter* $\alpha + \beta$ controlling how fast the data overtakes the prior.

Looking ahead: later chapters replace this simple parametric $\theta \in \Theta$ setting with a *universal* prior over all computable measures (Solomonoff's prior, related to Kolmogorov complexity), extending exactly the same Bayesian machinery to build the AIXI agent.